

Profit Farming Portal - Requirements Document

Farmer Data Capture, Adoption Readiness Tracking, and Representative Management Platform

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1. Purpose

This document defines the business and system requirements for the Profit Farming Portal. The portal will enable the organization to appoint multiple farmer representatives, capture structured farmer data in the field, and consolidate all records in a central database. Management users will then analyze this data using filters and reports to identify farmers with the highest potential to adopt organic or natural farming practices.

This document should be treated as the baseline for scope validation, delivery planning, and acceptance review.

2. Reference Context

The portal should align with national natural-farming program objectives while maintaining a strong operational focus on farmer profiling, representative productivity, and adoption-readiness reporting.

3. Data Repository Requirements - Dedicated Organizational Database

The portal must operate on a **dedicated centralized database** owned and controlled by the organization. All farmer records, representative activities, follow-up logs, and adoption scores must be stored in this controlled repository rather than in scattered files or temporary external sheets.

All farmer and operational data must remain in the organization's own managed repository.

- Single source of truth for farmer profiling, crop history, and conversion readiness.
- Representative-level ownership mapping for every captured record.

- Master data tables for crop, location hierarchy, farming type, and appointment status.
- Strict role-based access at data level (representative, manager, organic farming manager).
- Mandatory audit fields: created by, updated by, timestamps, and change history.

Repository Component	Minimum Requirement	Client Review Checkpoint
Farmer Master	Unique farmer profile with mobile/ID based de-duplication.	System blocks duplicate record creation and shows warning.
Representative Mapping	Every farmer record mapped to one active representative and location.	Manager can filter by representative and see exact allocation.
Activity Log	Each create/update/follow-up action logged with user and timestamp.	Audit trail visible for any selected farmer profile.
Document Store	Structured storage for attachments (images, PDFs, notes).	Attachments can be uploaded, previewed, and downloaded.
Backup Repository	Daily backup and monthly archival retention.	Backup log report available for admin review.

4. Business Objectives

- Digitally capture field-level farmer details through appointed representatives.
- Maintain a single, reliable database for all farmer records.
- Track representative-wise data collection performance.
- Identify and shortlist farmers who are most likely to adopt organic farming.
- After farmer conversion interest is confirmed, provide structured consulting support from Pruthashakti.
- Consulting scope must include irrigation planning, fertilizer planning, spray schedules, and crop advisory.
- Enable manager-level filtering and report generation for decision making.
- Enable the Organic Farming Manager to create and assign multiple representatives.

5. Online Data Storage and Management

- All data capture should occur online and synchronize with the central database in real time.
- Each representative should see only assigned villages/territories and related records.
- Managers should have live dashboards to monitor submissions and pending follow-ups.
- The storage model should support document and image attachments for field evidence and verification.
- Daily automated backups and retention policy should be mandatory.
- Validation rules should enforce data quality at entry time and minimize rework.

Online synchronization, backup control, and data-quality validation are mandatory for go-live approval.

Area	Detailed Requirement	Expected Evidence During Review
Availability	Portal should support 24x7 access for field users and managers.	Monthly uptime report shared by technical team.
Live Sync	Submitted form data should reflect in manager dashboard without manual import.	Farmer count updates immediately after save/submit.
Data Retention	No automatic deletion of active farmer records without approval workflow.	Delete action requires permission and maintains a soft-delete log.
Attachment Handling	Attachments should follow size/type rules and be linked to specific profiles.	Rejected upload message for invalid files and success logs for valid uploads.
Data Security	Data in transit and at rest should be protected using standard encryption.	Security configuration checklist reviewed and approved by the client.

6. Basic Data Processing Plan

The portal should provide an end-to-end data processing pipeline to convert raw field data into actionable farmer outreach plans.

Processing coverage must include extract, multi-layer sort, distribute, individual assignment, and executive overview.

Stage	Owner	Description	Service Level Target	Expected Outcome
Extract	Farmer Representative	Collect and standardize records from representative submissions.	Within same day of field visit	Validated primary dataset ready for analysis.
Multi-layer Sort	Farmer Response Manager	Sort by geography, crop type, land size, farming style, and adoption readiness score.	Daily processing run	Segmented farmer pools by priority and intervention type.
Distribute	Farmer Response Manager	Distribute segmented farmer lists to managers or field teams based on region or campaign objective.	Within 24 hours after sorting	Balanced workload and better campaign execution.
Assign to Individual	Organic Farming Manager	Assign targeted follow-up tasks to specific representatives with due dates.	Weekly assignment cycle	Clear accountability and measurable progress tracking.
Executive Overview	Management Team	Provide summary dashboards of overall data status, conversion funnel, and representative productivity.	Real-time dashboard and weekly summary	Leadership-level visibility for fast decision making.

7. User Roles and Access

Role	Key Responsibilities	Access Scope
Farmer Representative	Capture farmer profile, crop details, farming practices, and readiness indicators.	Role-specific credentials; create and update only assigned farmer records.
Farmer Response Manager	Monitor representative-wise data capture progress, filter data, generate adoption reports.	Read access to all farmer data, dashboards, and reporting modules.

Role	Key Responsibilities	Access Scope
Organic Farming Manager	Create representative accounts, assign territories, monitor outcomes, approve actions.	Full control of representative management and reporting.
System Admin (optional)	Platform setup, role policy, master data configuration.	Full system configuration and audit access.

8. Functional Requirements

FR-01: Authentication and Login

- Role-based login for each representative and manager.
- Secure password policy and account lock after repeated failed attempts.
- Optional OTP for manager-level login.

FR-02: Representative Onboarding

- Organic Farming Manager can create, edit, deactivate representative accounts.
- Representative assignment by village, cluster, taluka, or district.
- Track active and inactive representatives.

FR-03: Farmer Data Capture

- Representative captures complete farmer profile in structured forms.
- Save draft and submit workflow for incomplete field visits.
- Duplicate prevention by mobile number, Aadhaar or farmer ID.

FR-04: Central Database Management

- All submitted farmer data stored in a centralized database.
- Farmer records mapped with representative and geography.
- Time-stamped audit trail for create and update actions.

FR-05: Monitoring Dashboard

- Manager dashboard showing representative-wise data collection count.
- Daily, weekly, monthly trend of farmer registration.

FR-06: Filter and Report Engine

- Filter by location, crop type, land holding, irrigation, current farming method.
- Generate report for farmers likely to adopt organic farming.

- Pending vs completed profile records.

- Export reports in CSV and PDF formats.

FR ID	Detailed Requirement Statement	Priority	Acceptance Criteria
FR-01	System must provide role-based login and enforce module visibility based on role.	High	Representative cannot access manager reports; manager can access dashboard and reports only.
FR-02	Organic Farming Manager can create, edit, deactivate, and reassign representative accounts.	High	Account status changes reflect instantly and prevent unauthorized login.
FR-03	Farmer form must support draft save, final submit, and validation errors with field-level messages.	High	Incomplete mandatory fields block submit and show exact field name.
FR-04	System must maintain farmer history including updates, follow-ups, and readiness score changes.	High	Client can open one farmer profile and view complete timeline.
FR-05	Dashboard must display representative-wise capture count, conversion stage, and pending actions.	Medium	Dashboard reflects record counts correctly against filtered date range.
FR-06	Report module must support multilayer filtering and export to CSV/PDF.	High	Selected filters are applied in export output exactly as shown on the screen.
FR-07	Task assignment module must assign follow-up list to individuals with due date and status.	High	Assigned user receives task in the dashboard and can update completion status.
FR-08	System must provide a management bird's-eye view with summary KPIs and risk indicators.	Medium	Leadership view includes total records, high-potential farmers, pending follow-ups, and inactive representatives.

9. Farmer Data Fields (Minimum)

Category	Fields
Identity	Farmer Name, Mobile Number, Gender, Age, Farmer ID/Aadhaar (as per policy)
Location	Village, Taluka, District, State, Pincode, GPS coordinates (optional)
Farm Profile	Land Size, Irrigated/Non-irrigated area, Soil type, Water source
Cropping Details	Current crops, Seasonal pattern, Acre-wise crop coverage, Yield history
Farming Practice	Chemical/Organic/Natural farming status, Input usage, Current challenges
Adoption Readiness	Interest level, Awareness level, Training requirement, Conversion timeline preference
Engagement	Representative notes, Follow-up date, Appointment status, Next action
Consulting Plan	Irrigation (sichai) advisory, fertilizer plan, spray plan, crop-stage advisory, consultant assignment

Field Group	Mandatory Fields	Validation Rule
Identity	Farmer Name, Mobile Number	Mobile must be 10 digits and unique in active records.
Location	Village, Taluka, District	District must match selected state master list.
Farm Profile	Land Size, Water Source	Land size must be numeric and greater than zero.
Cropping Details	At least one current crop	Crop name must be selected from master catalog.
Adoption Readiness	Interest Level, Training Requirement	Interest level must be Low, Medium, or High.
Engagement	Follow-up Date for non-converted cases	Follow-up date cannot be earlier than current date.
Consulting Plan	Consultant Assignment, Consulting Start Date	Consulting Start Date must be on or after conversion confirmation date.

10. Adoption Scoring Logic (Initial)

The system should support a configurable readiness score to identify farmers with higher probability of early organic-farming adoption.

- Parameters include: interest level, current chemical dependency, water availability, training willingness, and past participation.
- Score bands: High (70-100), Medium (40-69), Low (0-39).
- Managers should be able to filter and generate reports by score band.

11. Key Reports

Report Name	Primary Filters	Consumer	Frequency
Representative Performance Report	Date Range, Representative, Village	Farmer Response Manager	Daily/Weekly
Location Coverage Report	District, Taluka, Village, Crop Type	Organic Farming Manager	Weekly
Adoption Readiness Funnel	Score Band, Crop Type, Land Size Band	Leadership Team	Weekly/Monthly
Pending Follow-up Register	Due Date, Assigned Representative, Status	Field Coordinator	Daily
Conversion Outcome Report	Date Range, Representative, Region, Conversion Stage	Management Team	Monthly
Post-Conversion Consulting Report	Consultant, Region, Crop, Irrigation Plan Status, Fertilizer Plan Status	Organic Farming Manager	Weekly/Monthly

12. Non-Functional Requirements

- Role-based access control with secure authentication.
- Mobile-friendly forms for field representatives.
- Data validation, duplicate checks, and error handling.
- Audit logs for critical actions.
- Backup and recovery process for central database.

- Average dashboard response time of under 3 seconds for standard filter combinations.

NFR Area	Target	Verification Method
Performance	Dashboard response under 3 seconds for standard filters.	Load test report and UAT validation.
Scalability	Support growth of records and users without redesign.	Database stress test summary.
Security	Strong authentication and access control by role.	Security checklist and permission audit.
Reliability	Daily backup with successful recovery drill.	Backup log and restore test evidence.
Usability	Field form completion within practical time window.	Pilot user feedback and completion time report.

13. Suggested Workflow

Step	Actor	Action	Outcome
1	Organic Farming Manager	Create and assign representative accounts.	Representatives receive login credentials.
2	Farmer Representative	Capture farmer details during field visits.	Data stored in central database.
3	Farmer Response Manager	Monitor captured data and representative performance.	Progress visibility by representative and geography.
4	Farmer Response Manager	Apply filters and generate adoption readiness reports.	Shortlisted list of farmers for conversion outreach.
5	Organic Farming Manager	Plan appointments and targeted campaign activities.	Higher conversion to organic farming adoption.

Step	Actor	Action	Outcome
6	Pruthashakti Consultant Team	Provide post-conversion advisory for irrigation (sichai), fertilizer, spray planning, and crop-specific practices.	Farmer receives end-to-end implementation support for organic farming.

14. Assumptions and Dependencies

- Final farmer form fields will be validated by business team before development.
- Regional language support may be added in a later phase.
- SMS and WhatsApp reminder integration can be planned as a Phase 2 enhancement.
- Data privacy and consent policy will be provided by management.

15. UAT Checklist for Client Review

UAT Item	Validation Scenario	Expected Result
Representative Login	Login with representative credentials.	Representative sees only assigned menus and farmer forms.
Farmer Record Creation	Create new farmer record with mandatory fields.	Record saved and visible in manager dashboard.
Duplicate Check	Try adding same mobile number again.	System blocks duplicate and shows alert.
Report Filtering	Apply location, crop, and readiness score filters.	Filtered results match selected criteria.
Task Assignment	Assign shortlist to one representative.	Assigned user can view tasks and update status.
Consulting Activation	Mark a converted farmer and assign consulting plan.	Consulting task list is generated with irrigation, fertilizer, spray, and crop advisory checkpoints.
Export Validation	Export filtered report in CSV/PDF.	Export content matches dashboard data.

16. Phase-Wise Delivery Suggestion

Phase	Focus	Outcome
Phase 1	Login, representative management, core farmer form, central repository.	Operational data capture baseline.
Phase 2	Dashboard, multilayer filters, report exports, assignment workflow.	Decision-ready processing and field task control.
Phase 3	Advanced scoring, communication integration, language enhancements.	Scalable conversion and engagement ecosystem.

17. Open Points for Client Confirmation

- Confirm final list of mandatory farmer fields and identity policy.
- Confirm readiness scoring formula and weightage per parameter.
- Confirm report templates and required export formats.
- Confirm assignment hierarchy between manager roles.
- Confirm data hosting preference and backup retention period.

Note: If any requirement does not align with the expected product outcome, please initiate a discussion for clarification before implementation.

End of document. This requirements draft is prepared for stakeholder review, scope confirmation, and implementation planning.